

Raw EEG Image Representation for CNN-Based Dementia Classification

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Abstract—EEG-based detection of Alzheimer’s disease (AD) and frontotemporal dementia (FTD) remains challenging under clinically realistic subject-disjoint evaluation. This study presents a lightweight convolutional neural network (CNN) framework using a raw EEG image representation that preserves temporal structure through normalized amplitude, first-order temporal difference, and squared-amplitude descriptors. The proposed representation is evaluated on a public OpenNeuro EEG dataset and compared with Discrete Wavelet Transform (DWT) time–frequency images under strict subject-wise 10-fold cross-validation to prevent subject-level data leakage. Across AD versus cognitively normal (CN), FTD versus CN, and combined AD+FTD versus CN tasks, the proposed raw EEG representation consistently outperformed DWT, achieving accuracies of 98.78%, 95.53%, and 97.04%, respectively, while also surpassing adapted EEG-specific CNN baselines. Grad-CAM and channel-wise analyses further show that the model exploits complementary discriminative information across amplitude, temporal variation, and energy-related descriptors. These results indicate that raw signal-preserving EEG representations provide an effective, interpretable, and computationally efficient alternative to transform-based inputs for CNN-based dementia classification.

Index Terms—electroencephalography, dementia classification, Alzheimer’s disease, frontotemporal dementia, convolutional neural networks, raw EEG image representation, time-frequency analysis, Grad-CAM.

I. INTRODUCTION

Dementia comprises progressive neurodegenerative disorders characterized by cognitive decline that interferes with daily functioning [1]. Alzheimer’s disease (AD) accounts for approximately 60–70% of cases, while frontotemporal dementia (FTD) is an earlier-onset subtype associated with behavioral and language impairments [2], [3]. Despite distinct neuropathologies, AD and FTD often exhibit overlapping clinical symptoms, complicating differential diagnosis [4]. Although MRI and PET provide valuable biomarkers, their cost and limited accessibility motivate complementary computational approaches for scalable dementia detection.

Electroencephalography (EEG) offers a practical alternative due to its non-invasive nature, relatively low cost, and high

temporal resolution for capturing neural dynamics associated with neurodegeneration [5], [6]. Convolutional neural networks (CNNs) enable automatic extraction of discriminative EEG patterns and have demonstrated promising performance for dementia classification [1], [7]. However, reported performance often depends strongly on signal representation, preprocessing choices, and evaluation design, underscoring the need for representations that generalize under clinically realistic subject-disjoint validation [8].

Many existing approaches employ time–frequency (TF) representations such as discrete wavelet transform (DWT) or spectrogram-based images to convert EEG signals into CNN-compatible inputs [1], [6]. While TF representations capture multi-resolution spectral structure, the transformation process may attenuate fine-grained temporal dynamics due to inherent TF resolution tradeoffs [9], [10]. In contrast, prior studies suggest that CNNs can learn discriminative structure directly from minimally processed EEG when temporal relationships are preserved [10]–[13]. Preserving signal-domain characteristics may therefore provide complementary information to TF-based approaches.

Motivated by these observations, this study investigates whether preserving raw EEG temporal structure improves CNN-based dementia classification relative to TF representations. We introduce a raw EEG image formulation that encodes normalized amplitude, first-order temporal difference, and squared amplitude as complementary descriptors compatible with standard 2D CNN architectures. The proposed representation is compared against DWT-based TF images using an ablation-optimized compact CNN and evaluated under strict subject-wise 10-fold cross-validation.

The methodological contributions are: (i) a raw EEG representation that encodes amplitude, temporal-difference, and energy descriptors for 2D CNN learning; (ii) a controlled DWT comparison using the same compact CNN; (iii) strict subject-wise validation with EEG-specific CNN benchmarks; and (iv) Grad-CAM and channel-wise analyses of learned descriptors.

TABLE I: Dataset characteristics.

Group	N	Males	Age	MMSE
AD	36	12	66.39 $\pm$ 7.89	17.75 $\pm$ 4.50
FTD	23	14	63.60 $\pm$ 8.20	22.17 $\pm$ 8.22
CN	29	18	67.90 $\pm$ 5.40	30.00 $\pm$ 0.00

II. METHODS

A. Dataset and Preprocessing

Experiments were conducted using a publicly available OpenNeuro EEG dataset comprising 88 participants: 36 with AD, 23 with FTD, and 29 cognitively normal (CN) individuals [14]. Table I summarizes the demographic and clinical characteristics.

EEG signals were band-pass filtered between 0.5–45 Hz and re-referenced. Artifact Subspace Reconstruction (ASR) [15] was applied using a 0.5 s sliding window with threshold 17 to suppress high-amplitude artifacts. Independent Component Analysis (ICA) [16] was then used to remove ocular and muscular components, and cleaned signals were reconstructed using neural sources only.

B. Discrete Wavelet Transform Representation

DWT-based time–frequency (TF) images were generated from EEG segments as the transform-based baseline for comparison with the proposed raw-signal formulation. DWT provides multi-resolution analysis by decomposing signals into localized TF components [17].

Given a signal $x(t)$, the DWT is defined as:

$$\text{DWT}(j, k) = \frac{1}{\sqrt{|2^j|}} \int_{-\infty}^{\infty} x(t) \psi\left(\frac{t - 2^j k}{2^j}\right) dt. \quad (1)$$

At each level, the decomposition produces approximation and detail coefficients corresponding to coarse and high-frequency signal components [18]. The resulting TF images serve as the DWT baseline shown in Fig. 1(a).

C. Proposed Raw-Signal Representation for 2D CNN Learning

We propose a raw EEG image representation that preserves temporal structure while remaining compatible with standard 2D CNN architectures. Each EEG segment $X \in \mathbb{R}^{C \times T}$ is transformed into three complementary signal-domain descriptors:

1) Normalized amplitude

$$X_R(c, t) = \frac{X(c, t) - \mu_c}{\sigma_c + \epsilon} \quad (2)$$

2) First-order temporal difference

$$X_G(c, t) = \begin{cases} 0, & t = 1 \\ X_R(c, t) - X_R(c, t - 1), & t > 1 \end{cases} \quad (3)$$

3) Squared amplitude (energy proxy)

$$X_B(c, t) = X_R(c, t)^2 \quad (4)$$

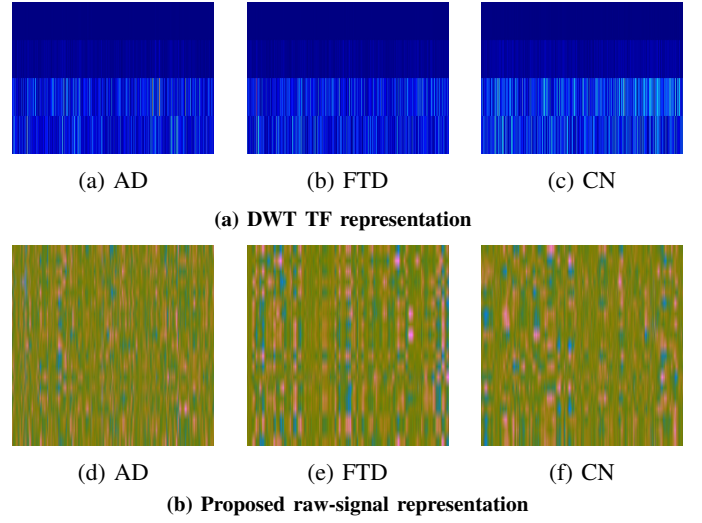


Fig. 1: Comparison of EEG image representations for CNN learning. Top: DWT TF images. Bottom: proposed raw-signal representation using amplitude, temporal-derivative, and energy channels.

These descriptors capture amplitude distribution, temporal variation, and instantaneous signal intensity. The matrices are stacked into a three-channel tensor:

$$X_{RGB}(c, t, :) = [X_R(c, t), X_G(c, t), X_B(c, t)]. \quad (5)$$

The tensor is resized to $224 \times 224 \times 3$ using bilinear interpolation to match CNN input dimensions. Examples of the proposed representation are shown in Fig. 1(b).

D. Architecture Selection and Optimization for 2D CNN Learning

The final architecture was obtained through ablation-driven optimization to identify a compact model for learning discriminative patterns from raw EEG image representations. Candidate models used a four-layer convolutional backbone with 3×3 kernels and progressively increasing filters (128, 256, 512, 512), each followed by Batch Normalization and ReLU activation.

Pooling configuration was the main design variable. A 2×2 MaxPooling layer after the second convolution reduced the feature maps from 224×224 to 112×112 while preserving intermediate spatial structure. A structured 16×16 Average-Pooling layer then produced compact 7×7 feature maps before flattening.

The classifier head consists of two fully connected layers with 256 and 64 units, followed by Dropout rates of 0.30 and 0.25, respectively. A Softmax output layer with $C = 2$ classes was used for binary classification. Training used AdamW with learning rate 1×10^{-4} and weight decay 5×10^{-5} .

E. Adapted EEG-Specific CNN Baselines

To benchmark the proposed representation, we evaluated established EEG-specific architectures originally developed

for raw EEG decoding, including EEGNet [19], EEGNet-SSVEP [20], DeepConvNet [11], and EEG-Inception [21]. These models represent common EEG learning strategies based on depthwise-separable convolutions, hierarchical convolution–pooling blocks, and multiscale parallel convolutions.

Because these architectures were originally designed for channel–time inputs, their input layers were adapted to accept $224 \times 224 \times 3$ images. Apart from this dimensional adjustment, architectural components were preserved as closely as possible to their original specifications. All baselines were trained using the same preprocessing pipeline and optimization settings as the proposed CNN to ensure a consistent comparison.

III. MODEL VALIDATION

Model robustness was evaluated using strict *subject-wise 10-fold cross-validation*, ensuring that all samples from each subject were assigned exclusively to either training or validation set within each fold.

In each fold, one subject per class was held out for validation, while the remaining subjects were used for training, such that all eligible subjects served once as validation cases.

Performance was measured using fold-wise accuracy and reported as mean $\pm$ standard deviation. Aggregated predictions across folds were used to compute precision, recall, and F1-score, providing stable estimates of subject-level generalization.

A. Model Interpretability via Grad-CAM

Model interpretability was examined using Gradient-weighted Class Activation Mapping (Grad-CAM) [22], which highlights input regions that contribute most strongly to class predictions. Grad-CAM computes gradients of the pre-softmax score y^c with respect to feature maps of the final convolutional layer.

Let A^k denote the k -th feature map. The importance weight α_k^c is obtained by applying global average pooling to the gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \quad (6)$$

where Z is the spatial size of the feature map. The localization map is computed as:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right). \quad (7)$$

The resulting heatmap is upsampled to the input resolution and overlaid on the raw EEG image to visualize discriminative spatiotemporal patterns learned by the CNN (Fig. 2).

IV. RESULTS

A. Performance Comparison of Raw EEG and DWT Representations for 2D CNN Learning

Table II compares the proposed raw EEG representation with DWT TF images using the same compact 2D CNN under

TABLE II: Subject-wise CNN performance comparison between raw EEG and DWT representations.

Representation	Classes	Mean Acc. (%)	Std (%)
Raw EEG	AD/CN	98.78	2.43
	FTD/CN	95.53	6.36
	AD+FTD/CN	97.04	3.21
DWT	AD/CN	80.61	6.46
	FTD/CN	84.43	5.52
	AD+FTD/CN	79.86	9.56

subject-wise 10-fold validation. Across all classification tasks (AD/CN, FTD/CN, AD+FTD/CN), raw EEG consistently outperformed DWT, indicating that preserving temporal signal structure provides a more discriminative feature space for subject-wise dementia classification.

For AD/CN, raw EEG achieved 98.78% (± 2.43) accuracy compared with 80.61% (± 6.46) for DWT, representing the largest performance gain. This suggests that signal-domain descriptors capturing amplitude, temporal variation, and local energy better characterize Alzheimer’s-related neural alterations than transform-based TF features.

For FTD/CN, raw EEG attained 95.53% (± 6.36), exceeding DWT performance of 84.43% (± 5.52). Although the margin was smaller than in AD/CN, the raw representation maintained a clear performance advantage, indicating improved sensitivity to heterogeneous temporal patterns associated with frontotemporal degeneration.

For the more challenging AD+FTD/CN task, raw EEG achieved 97.04% (± 3.21), substantially higher than DWT at 79.86% (± 9.56). Despite increased intra-class variability, the raw representation maintained both higher accuracy and lower variability, suggesting stronger generalization under strict subject-wise evaluation.

Consistent trends are observed in Table III, where macro-averaged precision, recall, and F1-score further confirm the advantage of raw EEG across all classification settings. For AD/CN, raw EEG achieved near-perfect performance with an F1-score of 98.77%, indicating highly balanced classification with minimal false positives and false negatives, while DWT showed substantially lower performance with an F1-score of 80.42%.

For FTD/CN, raw EEG maintained strong discriminative capability with an F1-score of 95.70%, compared with 84.70% for DWT. This reflects improved precision–recall balance despite the known heterogeneity of FTD patterns.

The largest disparity occurred for AD+FTD/CN, where DWT exhibited reduced recall of 62.62% and F1-score of 64.34%, suggesting difficulty identifying dementia cases under increased subtype variability. In contrast, raw EEG maintained stable performance with an F1-score of 96.89%, indicating greater robustness to inter-class variability.

Overall, these results demonstrate that the proposed raw EEG representation provides a more effective and stable input for 2D CNN learning than DWT-based TF images. While DWT offers a compact multi-resolution spectral summary, the transformation may attenuate fine-grained temporal structure that remains diagnostically informative. In contrast, the raw

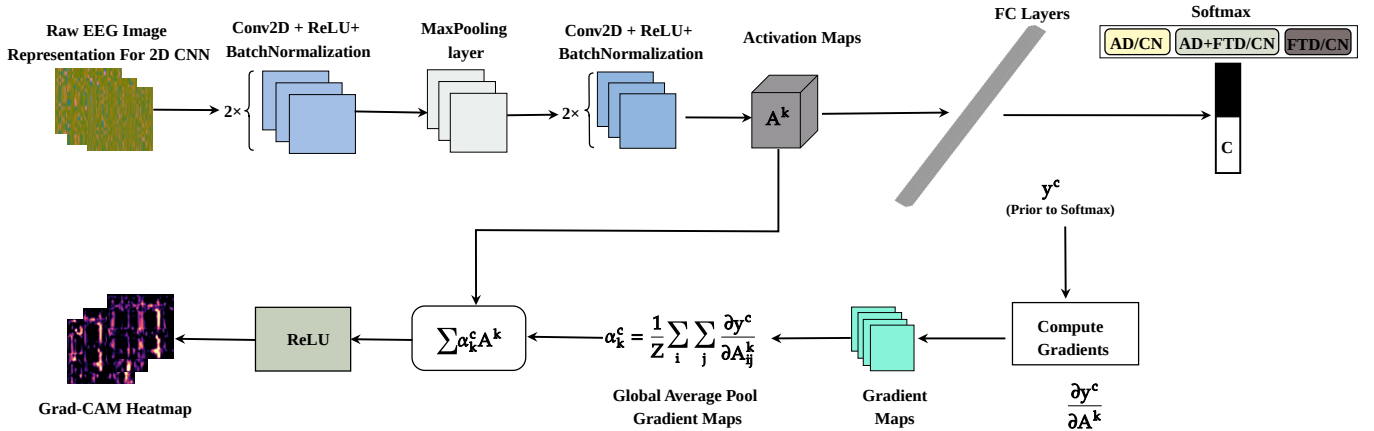


Fig. 2: Grad-CAM mechanism highlighting discriminative spatial patterns by propagating gradients from the target class to convolutional feature maps.

TABLE III: Subject-wise macro-averaged precision, recall, and F1-score.

Rep.	Classes	Prec. (%)	Rec. (%)	F1 (%)
Raw EEG	AD/CN	98.75	98.78	98.77
	FTD/CN	95.71	95.73	95.70
	AD+FTD/CN	97.18	96.61	96.89
DWT	AD/CN	80.51	80.46	80.42
	FTD/CN	84.82	84.66	84.70
	AD+FTD/CN	72.48	62.62	64.34

representation preserves richer signal-level dynamics, yielding consistently higher accuracy and improved robustness across classification tasks.

B. Grad-CAM Analysis of Learned EEG Discriminative Patterns

Figure 3 presents representative Grad-CAM heatmaps and corresponding overlay visualizations for the proposed raw EEG image representation. The heatmaps exhibit spatially localized and class-discriminative activation patterns, indicating that the CNN consistently attends to structured spatiotemporal signal regions when forming predictions. The overlay visualizations further show clear alignment between high-importance regions and salient amplitude–temporal dynamics within the raw EEG inputs, suggesting that the model captures meaningful neurophysiological structure rather than diffuse or unstable responses. These observations support the reliability of the learned representation under subject-wise evaluation.

C. Channel-Wise Contribution Analysis of the Raw EEG Representation

To further examine how the proposed representation supports discriminative learning, Fig. 4 illustrates channel-wise Grad-CAM contribution maps for normalized amplitude, temporal derivative, and signal energy components. The results show that discriminative evidence is distributed across all three channels, confirming that the CNN exploits complementary signal characteristics rather than relying on a single descriptor.

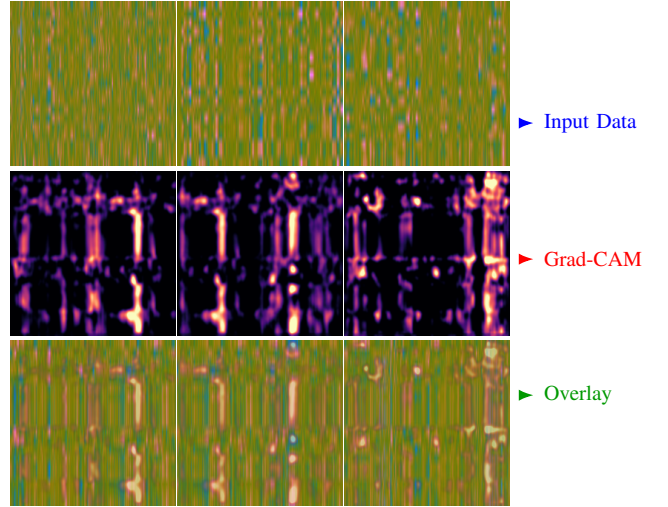


Fig. 3: Grad-CAM heatmaps and overlays for representative raw EEG samples.

The amplitude channel emphasizes global signal morphology and sustained intensity structure, while the derivative channel produces localized responses related to dynamic transitions and short-term waveform variation. The energy channel shows consistent activations, suggesting that local signal magnitude provides additional discriminative cues. These complementary patterns are consistent with Section II-C and indicate that performance gains arise from the combined exploitation of amplitude, temporal variation, and energy attributes.

D. Comparison with EEG-Specific Architectures

Figure 5 compares the proposed raw EEG 2D CNN with established EEG-specific architectures using the proposed raw EEG representation as input.

For AD/CN classification, the proposed model achieved 98.78% accuracy ($\pm 2.43\%$), with a clear margin over all EEG-specific baselines. The higher accuracy and markedly lower fold-level variability indicate improved robustness to inter-subject variability.

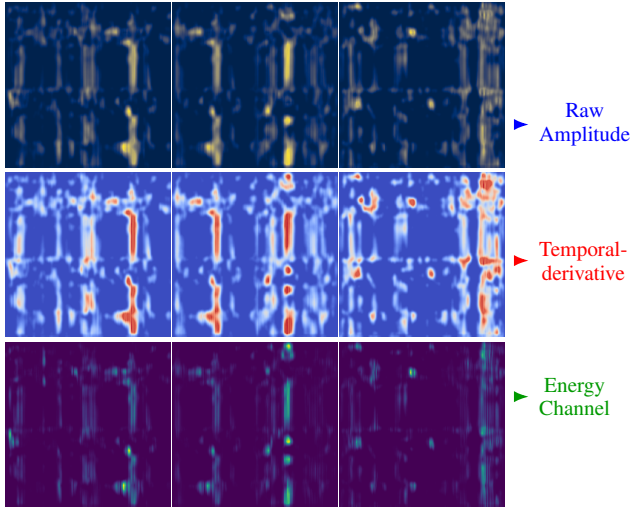


Fig. 4: Channel-wise Grad-CAM contribution maps for amplitude, temporal derivative, and energy components.

A similar trend was observed for FTD/CN, where the proposed CNN achieved 95.53% accuracy ($\pm 6.36\%$) and remained above all baseline models. The reduced dispersion across folds suggests more stable generalization across heterogeneous FTD presentations.

For the pooled AD+FTD/CN task, the proposed model reached 97.04% accuracy ($\pm 3.21\%$), again outperforming the EEG-specific baselines. This consistent advantage across tasks indicates that the proposed representation effectively captures discriminative neural dynamics shared across dementia subtypes.

Overall, the results suggest that architectures originally designed for temporal EEG decoding may not fully exploit structured signal representations derived directly from raw EEG. In contrast, the proposed CNN demonstrates stronger subject-level generalization and improved stability, highlighting the importance of architecture design and hyperparameter tuning when using the proposed input representation.

V. DISCUSSION

This study presents a lightweight 2D CNN framework using a raw EEG image representation that preserves temporal structure through normalized amplitude, temporal derivative, and signal energy channels. Under strict subject-wise cross-validation, the proposed representation consistently outperformed the DWT-based TF representation across all tasks, achieving higher accuracy and lower variability. These findings suggest that direct signal-domain encoding preserves discriminative temporal dynamics that may be partially attenuated by spectral transformations, leading to improved subject-level generalization.

Performance differences between AD and FTD classification tasks align with prior findings that FTD exhibits greater heterogeneity and overlapping EEG characteristics, typically reducing separability relative to AD [1]. Grad-CAM and channel-wise analyses further show that the CNN leverages

TABLE IV: Comparison with prior studies using the same EEG dataset.

Study	Year	Methodology	Performance	Limitations
Velichko et al. [23]	2023	NNetEn features + SVM	ACC=88.45% (AD/CN)	k-fold (epoch-level)
Chen et al. [7]	2023	Dual-branch ViT + CNN	ACC=80.23% (AD/FTD/CN)	k-fold (epoch-level)
Rostamikia et al. [6]	2024	TF-domain features + SVM	ACC=93.5% (AD+FTD/CN); ACC=87.5% (AD/FTD)	k-fold (epoch-level)
Wang et al. [4]	2024	Aperiodic exponent + offset	AUC=0.73 (AD/FTD)	k-fold

complementary information across amplitude, temporal variation, and signal energy, supporting the effectiveness of multi-descriptor signal encoding for capturing meaningful neurophysiological structure.

Benchmark comparisons with studies using the same dataset (Table IV) further contextualize the proposed framework. Prior work based on entropy measures [23], spectral descriptors [4], [6], and hybrid deep learning models [7] often employs epoch-level k-fold validation, which may introduce optimistic bias due to sample-level overlap. In contrast, the proposed framework demonstrates strong performance under strict subject-wise validation and consistently outperforms EEG-specific architectures originally designed for channel-time inputs, suggesting that raw signal-preserving CNN representations provide a more clinically realistic, effective, and interpretable alternative to handcrafted spectral features and complex hybrid architectures for EEG-based dementia classification.

VI. CONCLUSION

This study introduced a raw EEG image representation for 2D CNN-based dementia classification under strict subject-wise cross-validation. By encoding normalized amplitude, temporal derivative, and signal energy as complementary channels, the proposed formulation preserves signal-domain dynamics while remaining compatible with a lightweight CNN. Across AD/CN, FTD/CN, and AD+FTD/CN tasks, the proposed approach outperformed DWT-based representations and adapted EEG-specific CNN models, achieving higher accuracy and improved fold-level stability.

Grad-CAM and channel-wise analyses show that the network learns structured patterns distributed across amplitude, temporal variation, and signal energy components, supporting a discriminative and interpretable feature space without complex feature engineering or multi-branch architectures. Overall, the results support raw EEG image representations as an effective and computationally efficient framework for EEG-based dementia detection. Future work will evaluate generalization on independent datasets, extend to multiclass disease staging, and integrate complementary neurophysiological or clinical biomarkers.

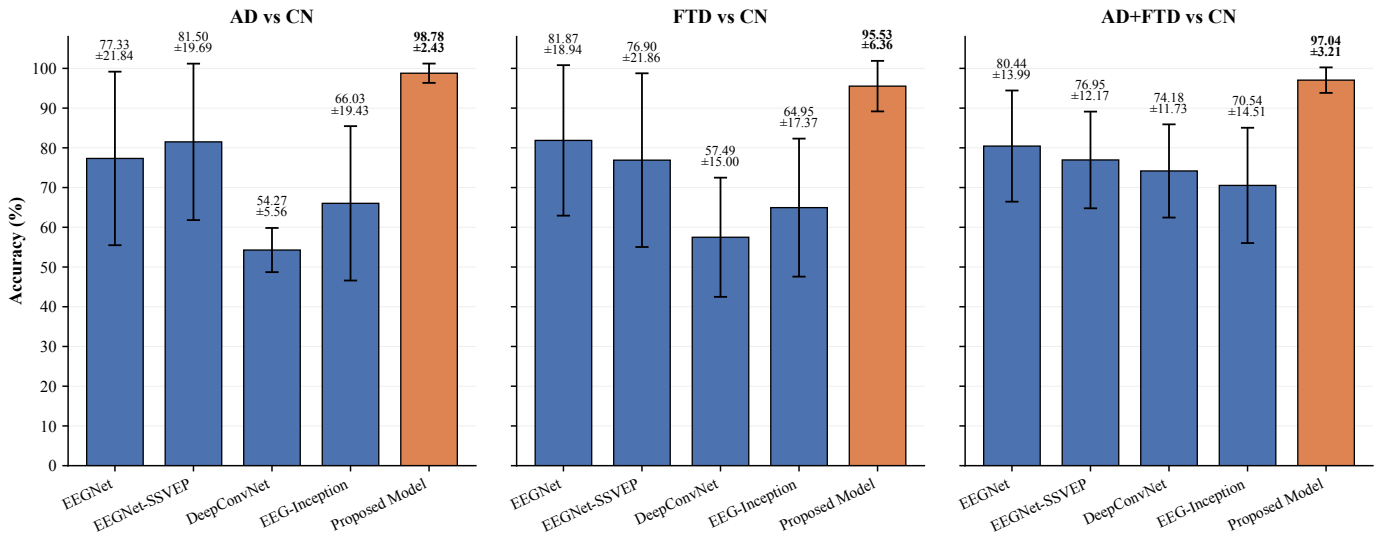


Fig. 5: Performance comparison of the proposed model and EEG-specific baselines using the raw EEG representation.

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